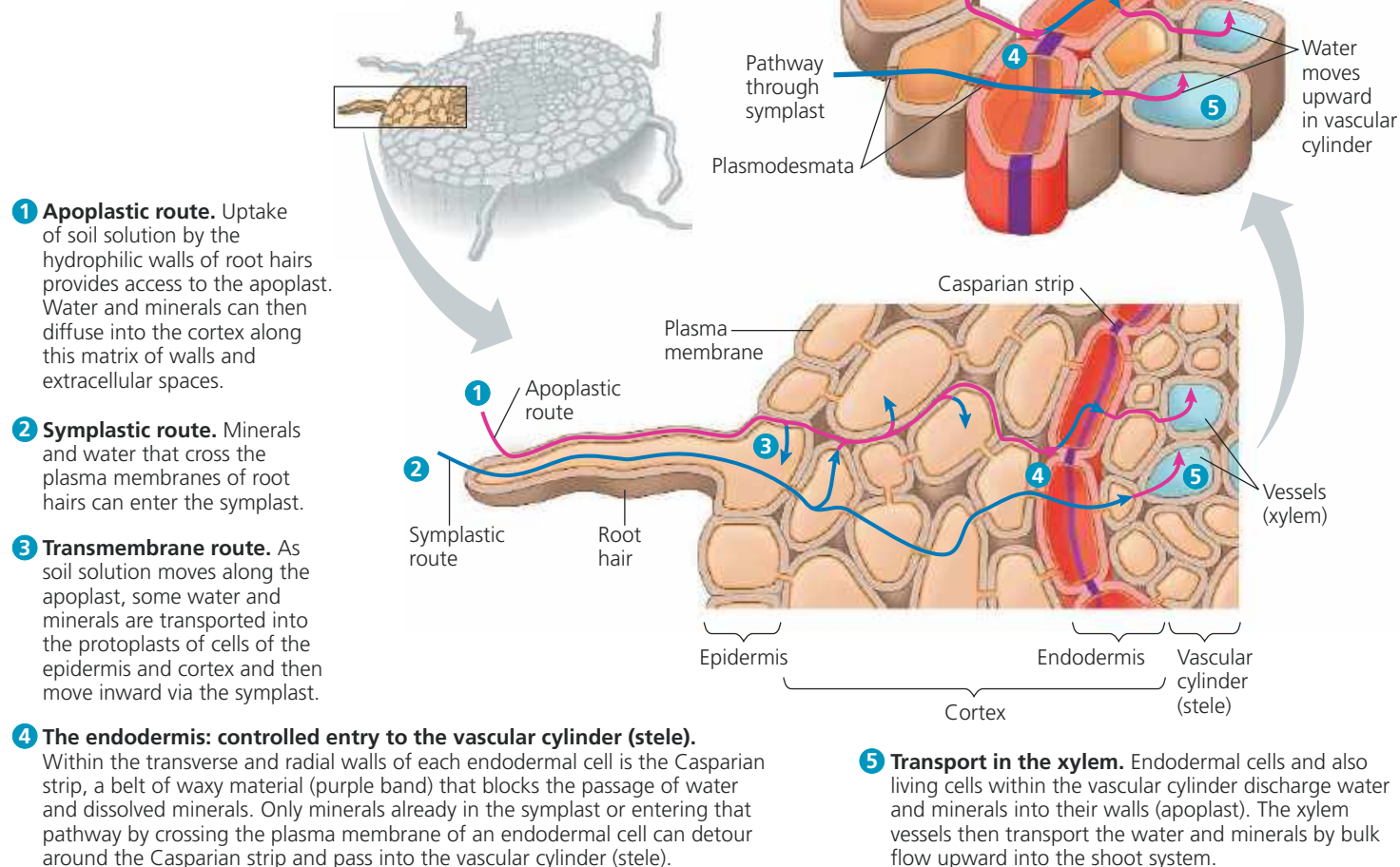


▼ **Figure 36.9** Transport of water and minerals from root hairs to the xylem.

VISUAL SKILLS After studying the figure, explain how the Casparian strip forces water and minerals to pass through the plasma membranes of endodermal cells.

➔ **Mastering Biology Figure Walkthrough**
BioFlix® Animation: Water Transport from Soil into Roots



A maize crop growing at a typical density of 60,000 plants per hectare transpires almost 4 million L of water per hectare (about 400,000 gallons of water per acre) every growing season. If the transpired water is not replaced by water transported up from the roots, the leaves will wilt, and the plants will eventually die.

Xylem sap rises to heights of more than 120 m in the tallest trees. Is the sap mainly *pushed* upward from the roots, or is it mainly *pulled* upward? Let's evaluate the relative contributions of these two mechanisms.

Pushing Xylem Sap: Root Pressure

At night, when there is almost no transpiration, root cells continue actively pumping mineral ions into the xylem of the vascular cylinder. Meanwhile, the Casparian strip of the endodermis prevents the ions from leaking back out into the cortex and soil. The resulting accumulation of minerals lowers the water potential within the vascular cylinder.

Water flows in from the root cortex, generating **root pressure**, a push of xylem sap. The root pressure sometimes causes more water to enter the leaves than is transpired, resulting in **guttation**, the exudation of water droplets that can be seen in the morning on the tips or edges of some plant leaves (**Figure 36.10**). Guttation fluid should not be confused with dew, which is condensed atmospheric moisture.

► **Figure 36.10**
Guttation. Root pressure is forcing excess water from this strawberry leaf.

